Probability and Statistic (for CS Students)

Lecture 9: Bernoulli and Binomial Random Variables

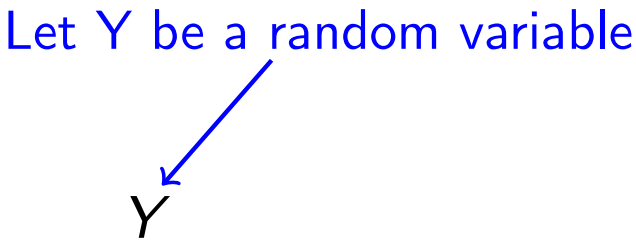
University of Mazandaran

Fall 1405

Review

- A random variable is a variable which takes on values probabilistically
- A discrete random variable is fully described by a probability mass function
- We can also calculate summary statistics such as expectation (and today, variance!)

Review



For example Y is the number of heads in 5 coin flips

$$Y = 2$$

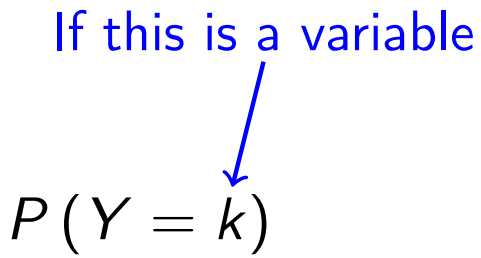
It is an event when Y takes on a value

For example Y is the number of heads in 5 coin flips

$$Y < 2$$

It is also an event when you ask any comparison question

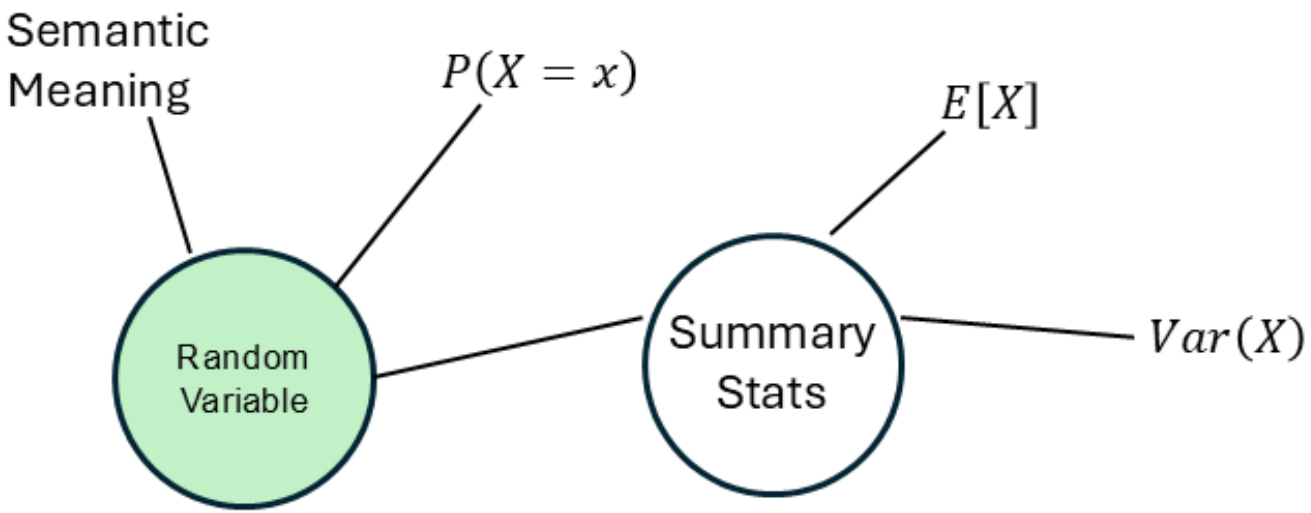
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Then this is a function

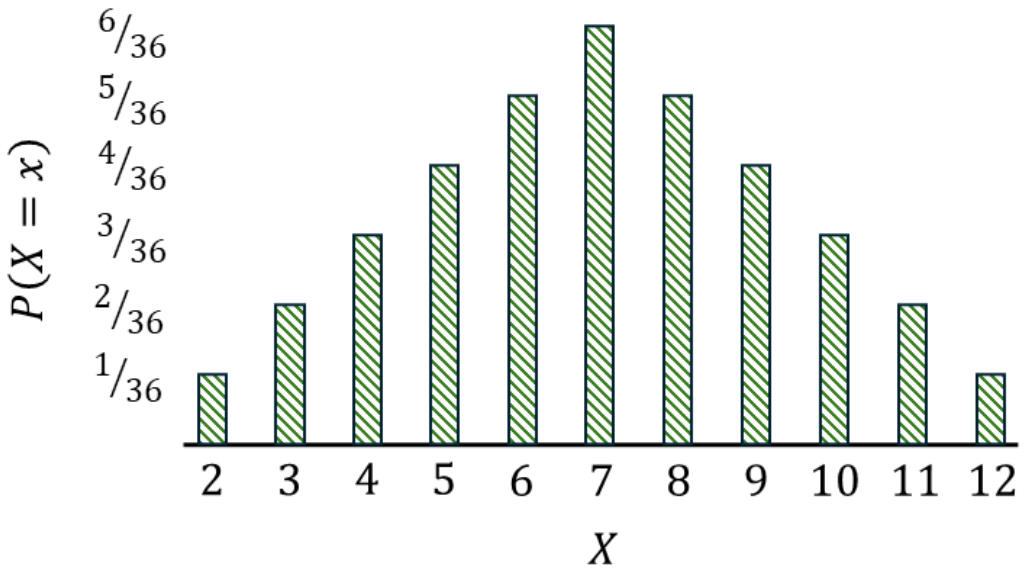
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Review

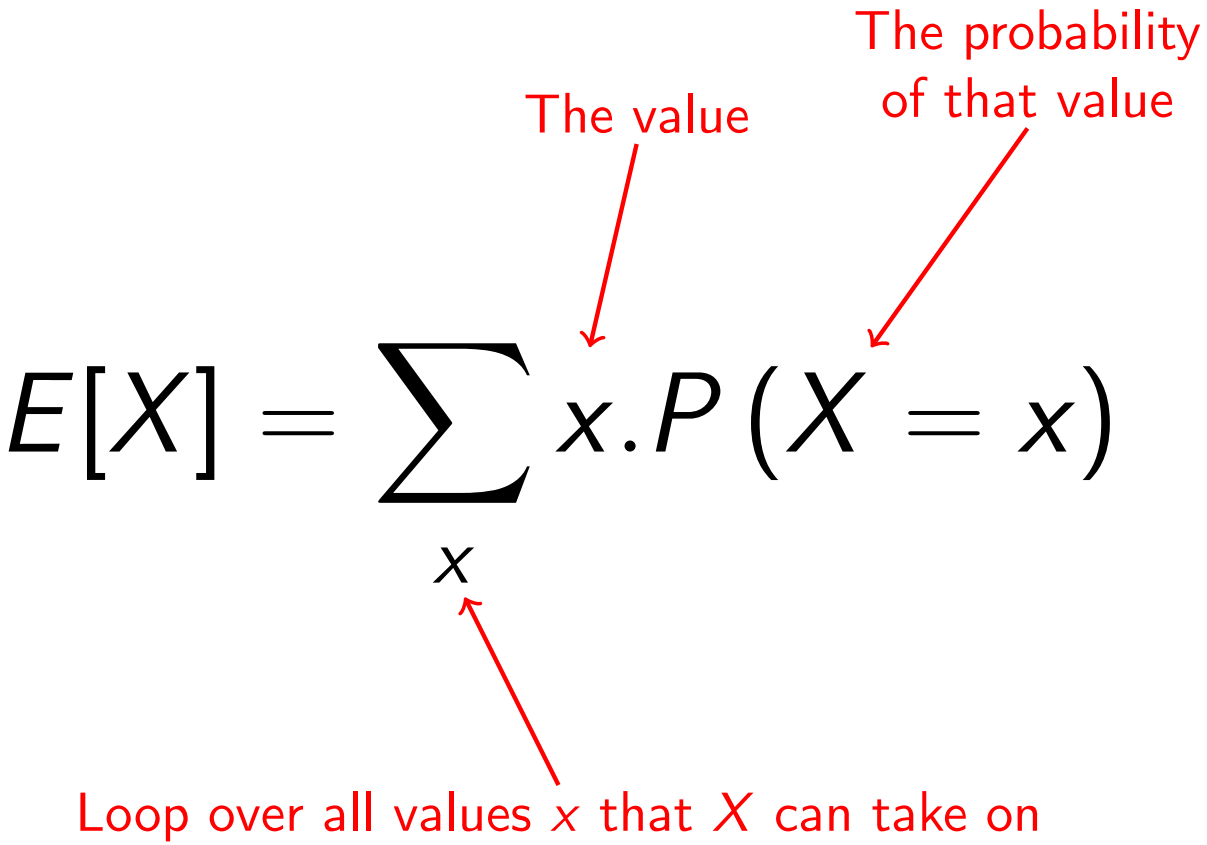

Semantic
Meaning
$P(X = x)$
$E[X]$
Random
Variable
Summary
Stats
$Var(X)$

PMF of sum of two dice

$$P(X = x) = \begin{cases} \frac{x-1}{36} & \text{if } x \in \mathbb{Z}, 1 \leq x \leq 6 \\ \frac{13-x}{36} & \text{if } x \in \mathbb{Z}, 7 \leq x \leq 12 \\ 0 & \text{else} \end{cases}$$



Expected Values



Is Expectation Enough?

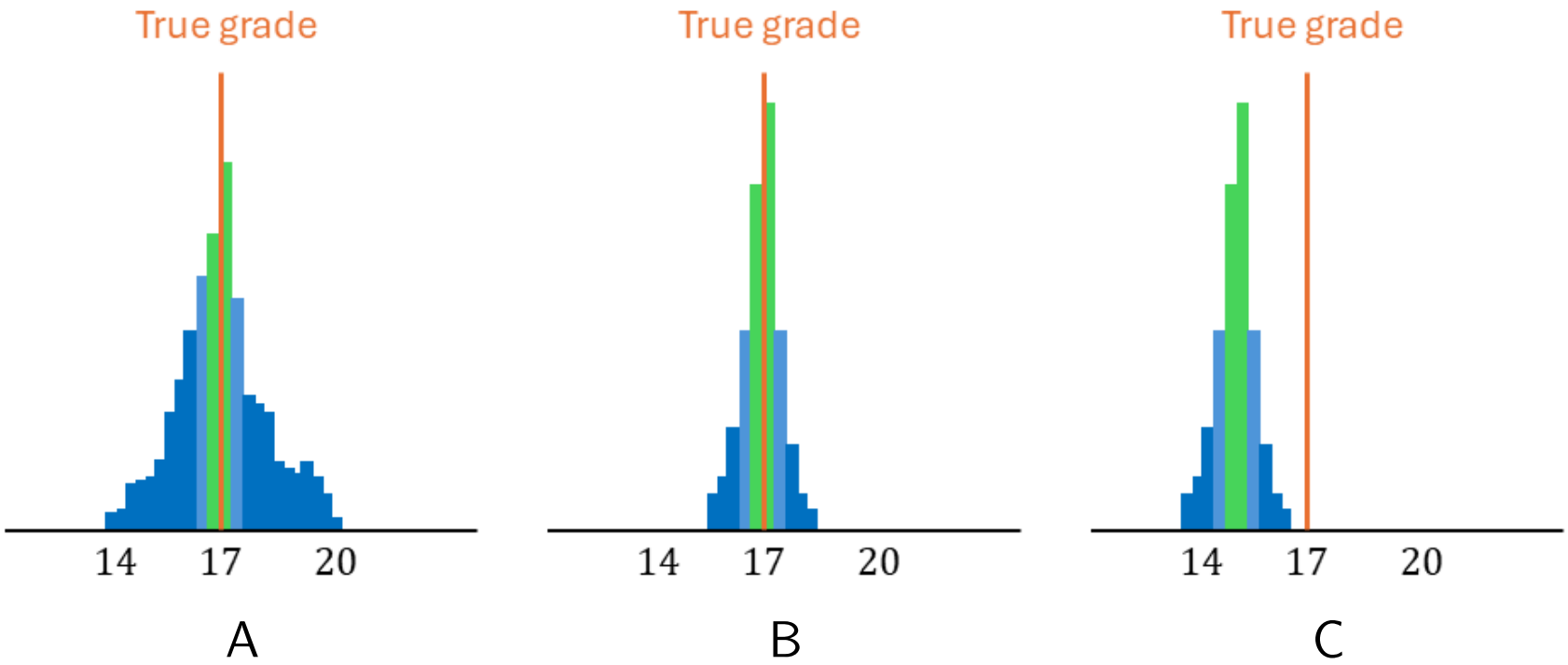
Is Expectation Enough?

No! PMF is complete!

Peer Grading

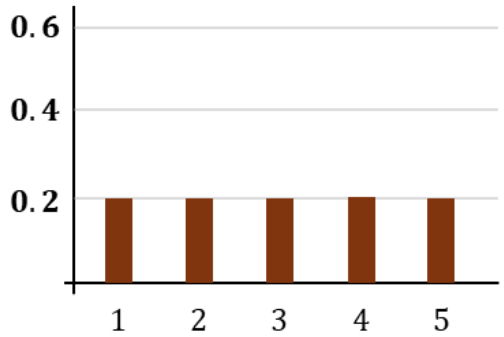
Peer grading on Coursera HCI
31067 peer grades for 3607 students.

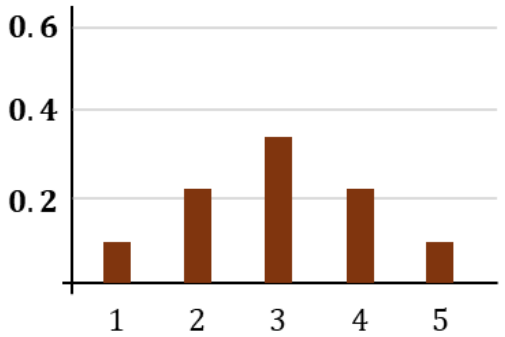
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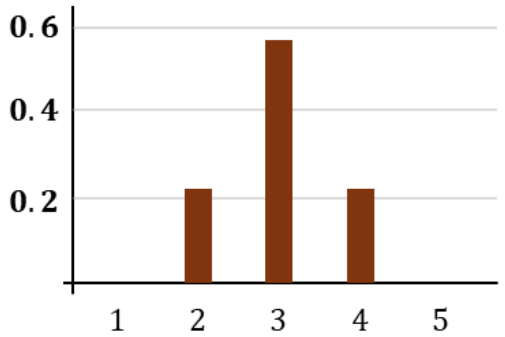


Measure of Spread

Consider the following 3 distributions (PMFs)

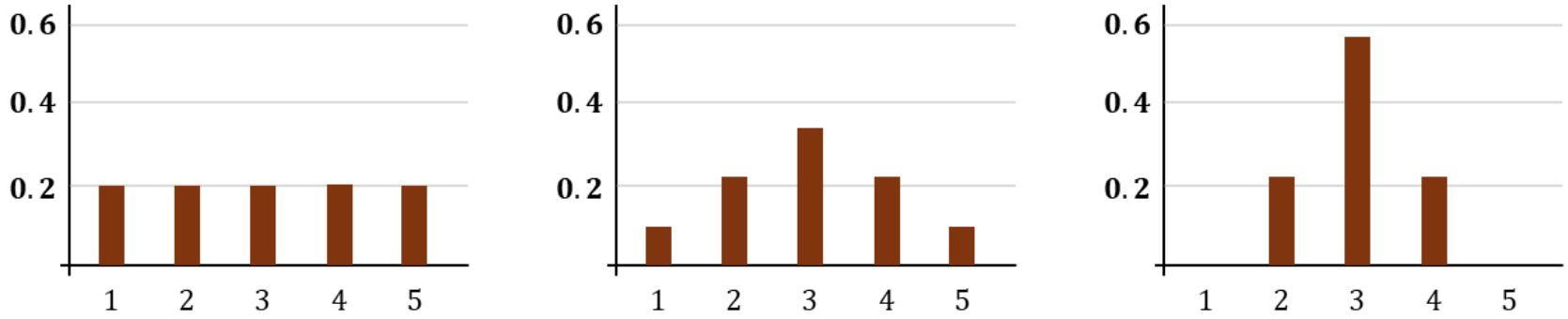






Measure of Spread

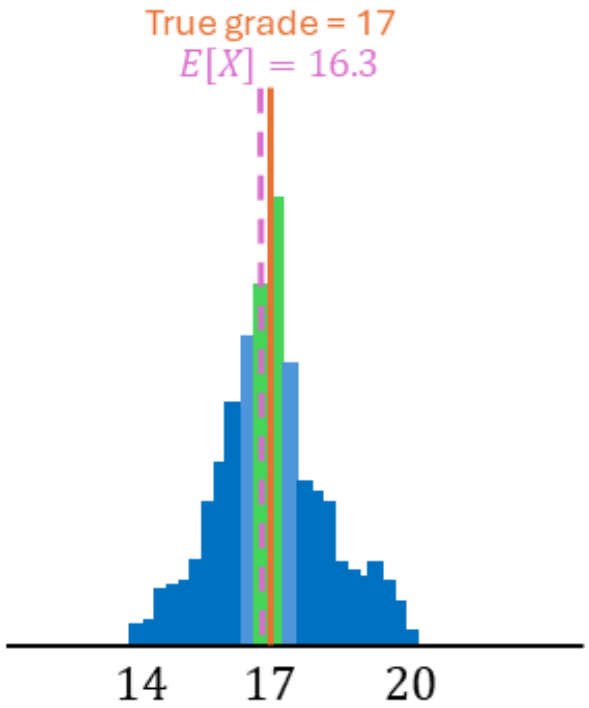
Consider the following 3 distributions (PMFs)



All have the same expected value, $E[X] = 3$
But "spread" in distributions is different
Invent a formal quantification of "spread"

Peer Grading in Coursera HCI

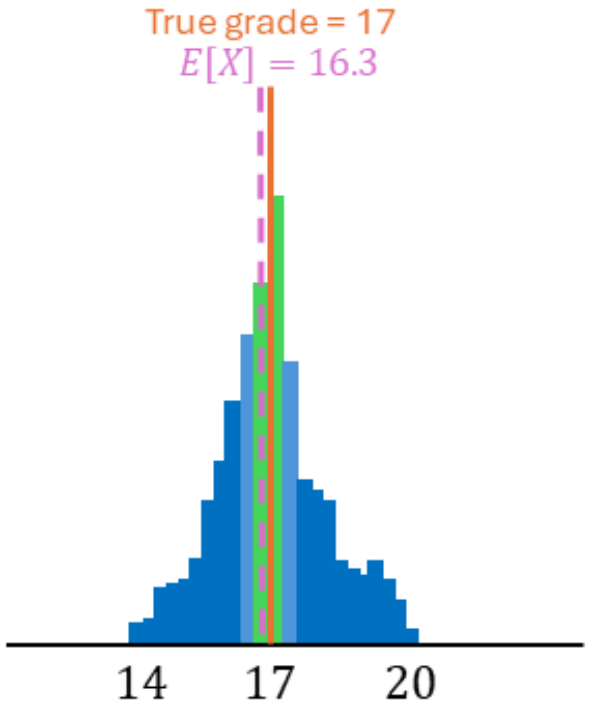
Let X be a random variable that represents a peer grade



Peer Grading in Coursera HCI

Let X be a random variable that represents a peer grade

$$\text{Var}(X) = E[(X - E[X])^2]$$



Variance

if X is a random variable with mean μ, then the variance of X, denoted by $\mathrm{Var}(X)$, is:

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Variance is a formal definition of the spread of a random variable

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Variance is a formal definition of the spread of a random variable

Also known as the 2nd moment, or square of the standard deviation.

Computing Variance

$$
\begin{aligned}
\mathrm{Var}(X) &= E[(X-\mu)^2] \\
&= \sum_x (x-\mu)^2 p(x) \\
&= \sum_x \left(x^2 - 2\mu x + \mu^2\right) p(x) \\
&= \sum_x x^2 p(x) - 2\mu \sum_x x p(x) + \mu^2 \sum_x p(x) \\
&= E[X^2] - 2\mu E[x] + \mu^2 \\
&= E[X^2] - 2\mu^2 + \mu^2 \\
&= E[X^2] - \mu^2 \\
&= E[X^2] - (E[X])^2
\end{aligned}
$$

Example

UMZ wants to equip its classrooms with interactive whiteboards. Two contractors are among the options and their monthly supply costs over previous 6 months are:

- **Contractor A**: \$960, \$1030, \$980, \$1000, \$1040, \$990
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Compute the mean and variance of the monthly cost for each contractor and suggest one to the dean?

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Compute the mean and variance of the monthly cost for each contractor and suggest one to the dean?

$$E[A] = \sum_{i=1}^{6} x_i p(x_i) = (960 + 1030 + 980 + 1000 + 1040 + 990) \times \frac{1}{6} = 1000$$

$$E[B] = \sum_{i=1}^{6} x_i p(x_i) = (500 + 1300 + 900 + 600 + 1500 + 1200) \times \frac{1}{6} = 1000$$

$$Var[A] = E\left[(A - E[A])^2\right] = 766.66$$

$$Var[B] = E\left[(B - E[B])^2\right] = 133333.3$$

Bernoulli Random Variable

Experiment results in “Success" or “Failure"

- X is random indicator variable (1 =success, 0 =failure)

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parameter: success probability

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- $E[X] = p$
- $Var(X) = p(1 - p)$

Again Coins!

1. n independent trials of a Bernoulli experiment (eg flipping a coin)
2. Each trial has a probability of p, of being a success (eg a head)
3. What is the probability of exactly k successes? (eg k heads)

Many Random Variables Follow This Pattern

Examples:

- # of heads in n coin flips
- # of 1's in randomly generated bit string of length n
- # of disk drives crashed in 1000 computer cluster
- # of people who vote for a candidate

Exactly k heads in n coin flips

X: R.V. that models the number of heads in n coin flips
Probability of exactly k heads:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

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X is called a binomial random variable

Declare a Random Variable to be Binomial

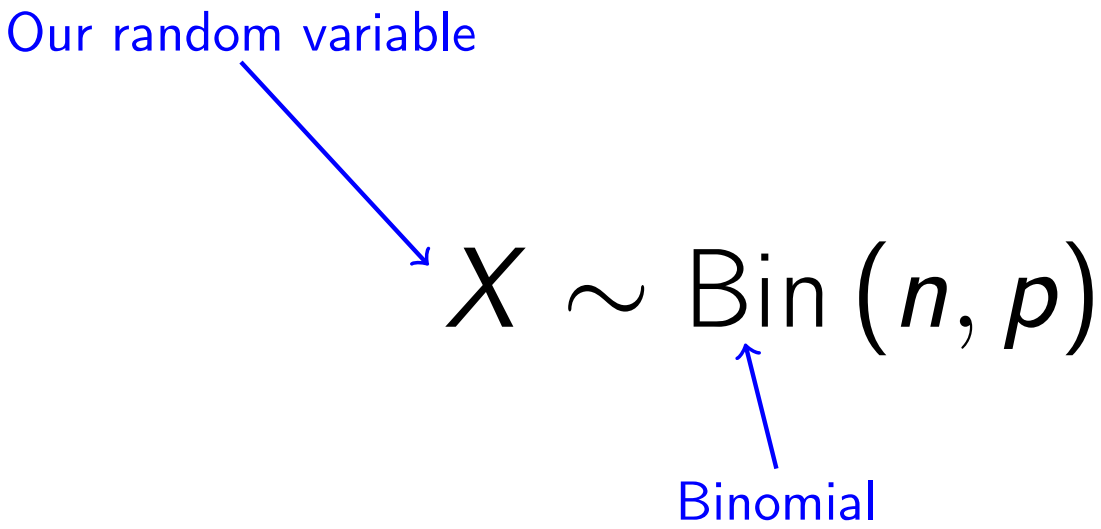
$$X \sim \text{Bin}(n, p)$$

Declare a Random Variable to be Binomial

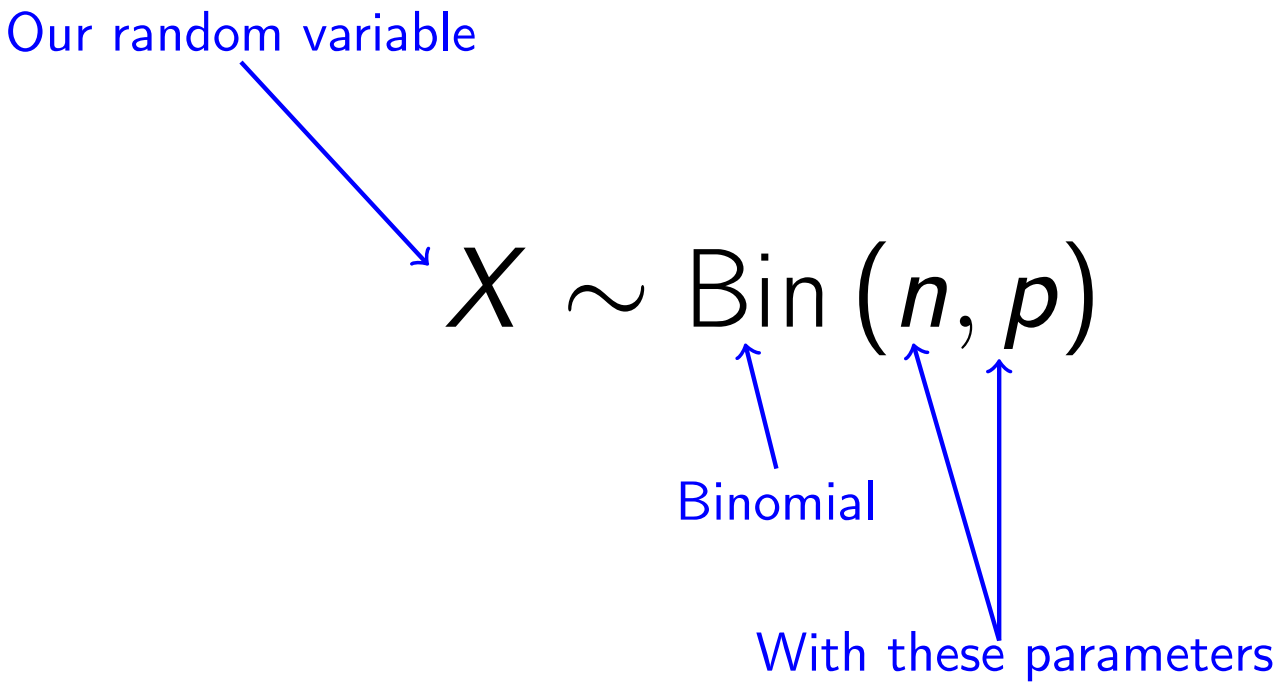
Our random variable

$$X \sim \text{Bin}(n, p)$$

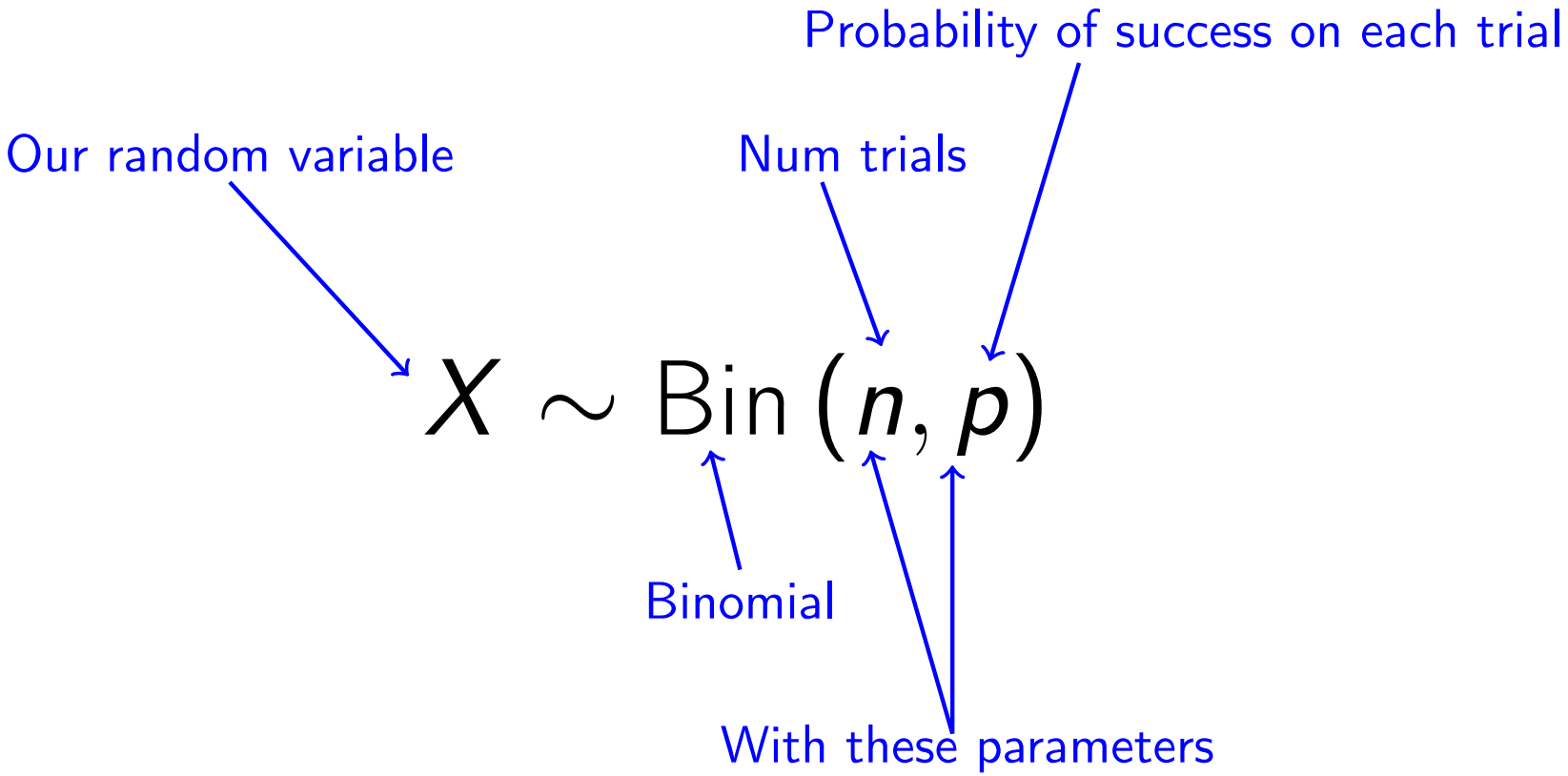
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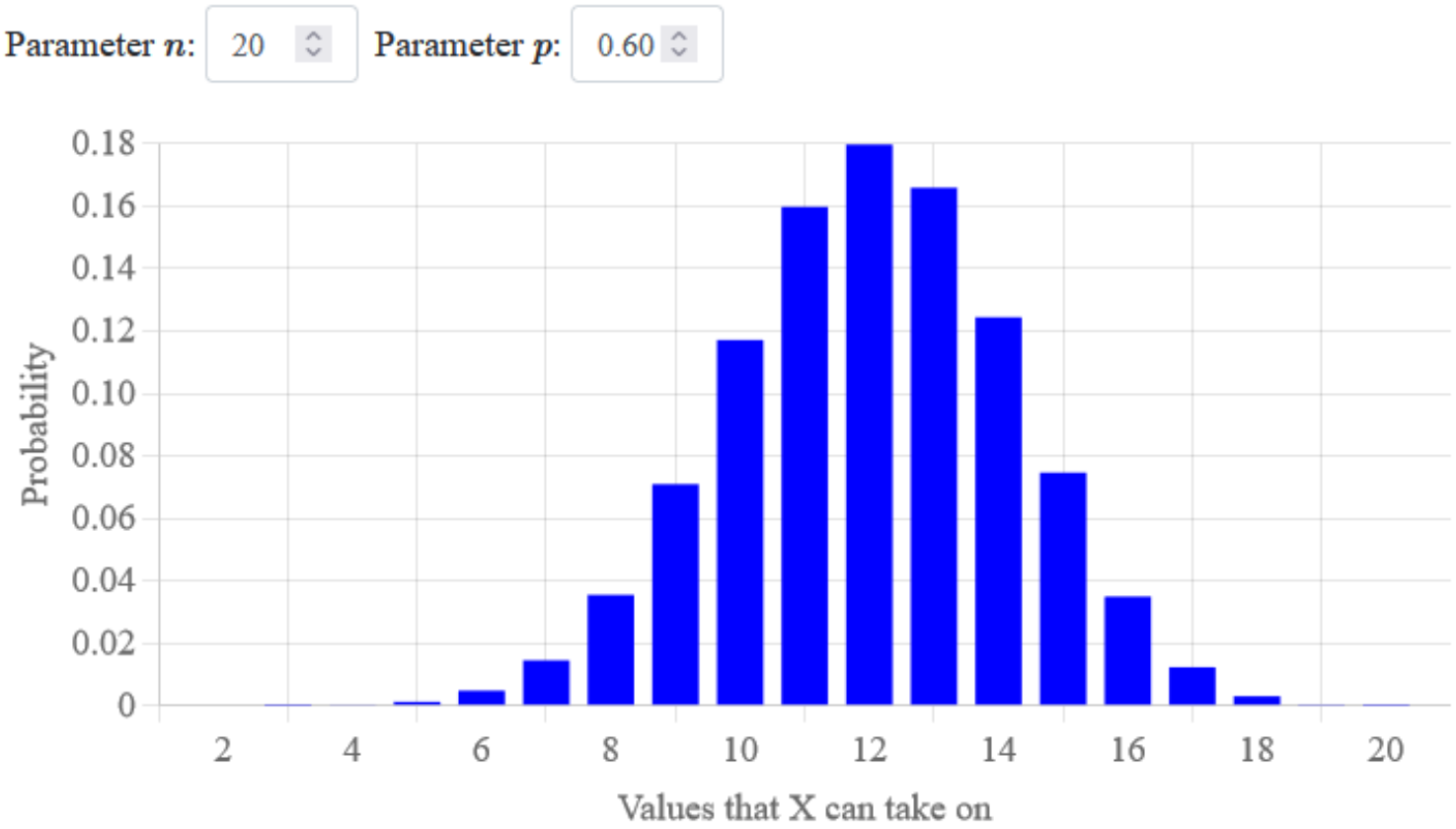
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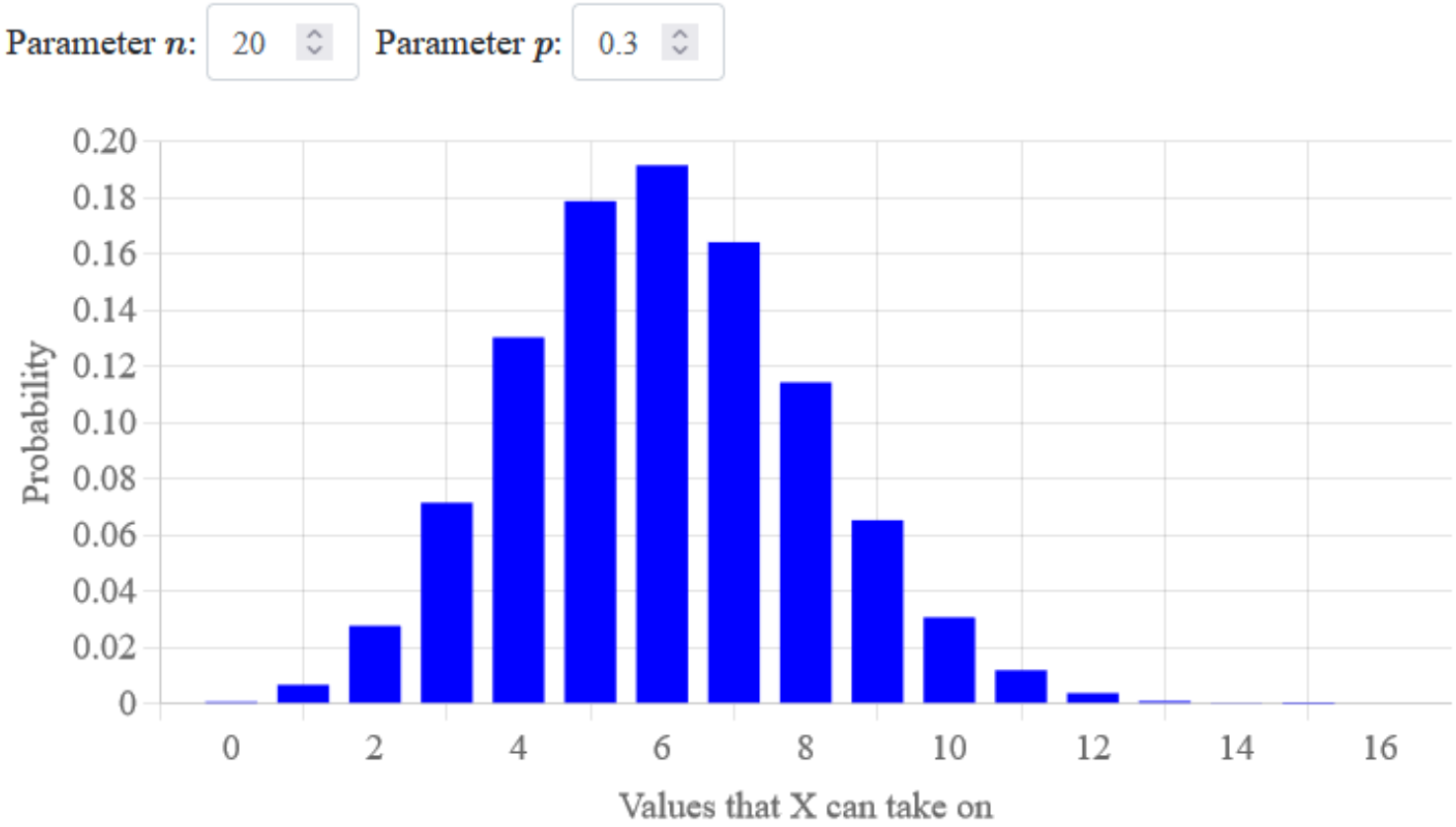
Binomial PMF

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

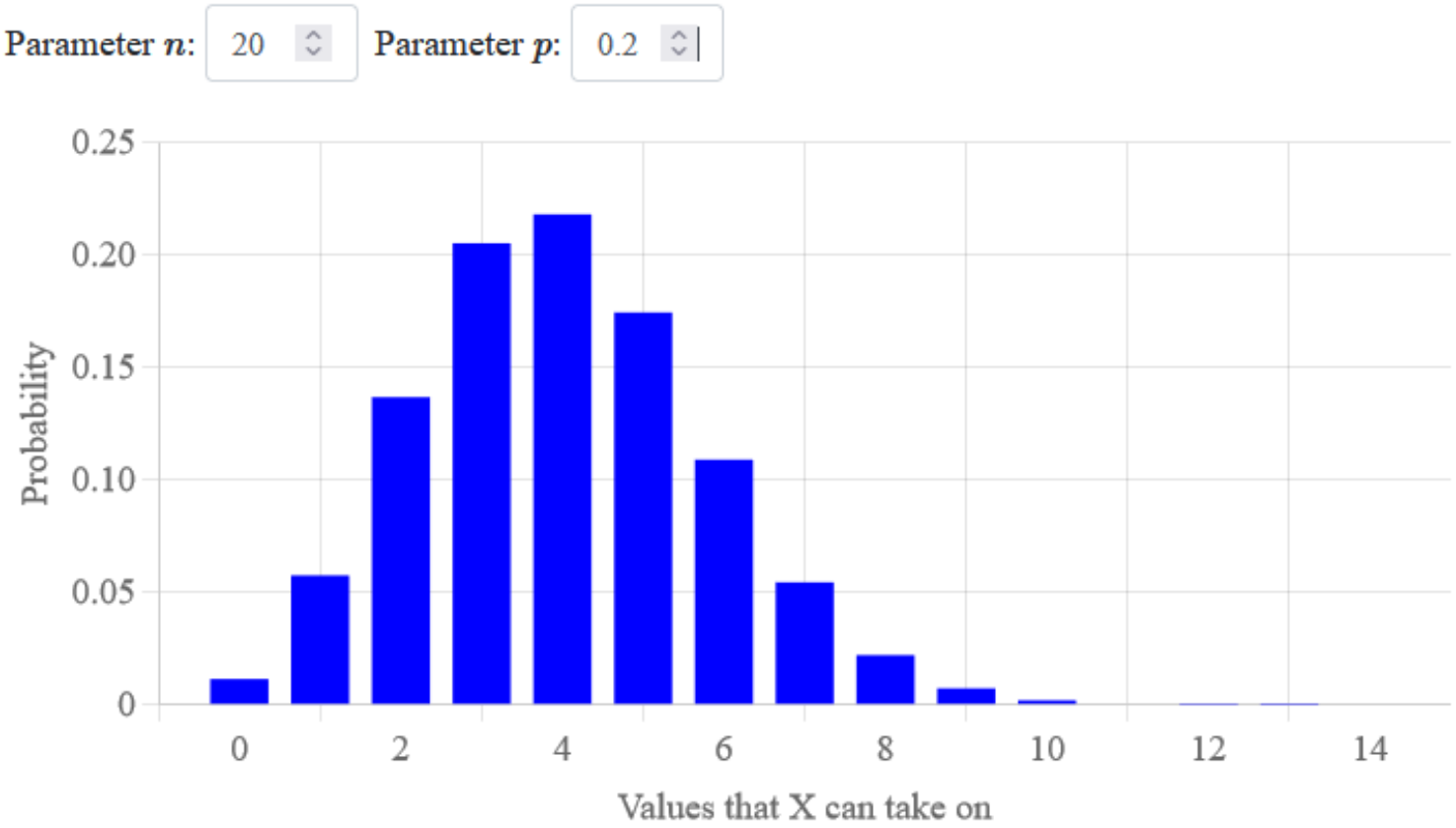
Binomial PMF



Binomial PMF



Binomial PMF



Coins, now with Binomial

Three fair ("heads" with $p = 0.5$) coins are flipped

- X is number of heads
- $X \sim Bin\,(n = 3, p = 0.5)$

$$P\,(X = 0) = \binom{3}{0} p^0\,(1-p)^3 = \frac{1}{8}$$

$$P\,(X = 1) = \binom{3}{1} p^1\,(1-p)^2 = \frac{3}{8}$$

$$P\,(X = 2) = \binom{3}{2} p^2\,(1-p)^1 = \frac{3}{8}$$

$$P\,(X = 3) = \binom{3}{3} p^3\,(1-p)^0 = \frac{1}{8}$$